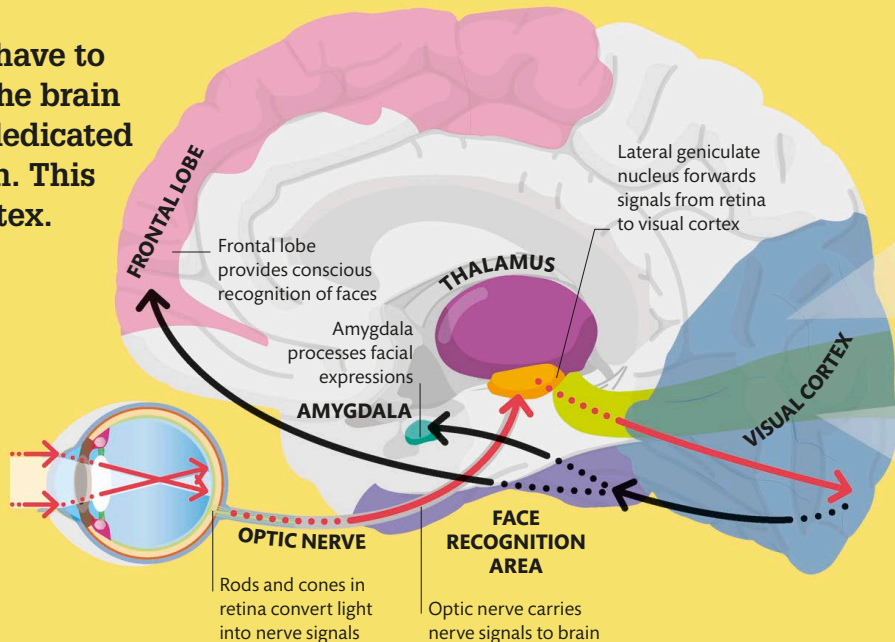


The Visual Cortex

Nerve signals from the eye have to travel all the way through the brain before they reach the area dedicated to decoding this information. This area is called the visual cortex.

The structure of the cortex

The visual cortex occurs in both brain hemispheres and is further divided into eight main areas, each of which has a different function (see table, opposite). Signals travel from the retina (see pp.66–67) via the thalamus and lateral geniculate nucleus to the primary visual cortex (V1). The raw data then passes through various vision areas, contributing different details about shape, color, depth, and motion before combining to form an image. Some areas provide information that helps with immediate recognition of familiar objects, others with spatial orientation or visual-motor skills.



1 From eyeball to visual cortex
Data from the eyeball travels along the optic nerve until it reaches the optic chiasm (see below), where some of the data is sent to the opposite side of the brain. Signals then travel to the lateral geniculate nucleus, which forwards data to the visual cortex for processing.

KEY
• → Information from the eye
• → Face recognition pathway

Stereoscopic vision

Our ability to see in 3-D—known as stereoscopic vision—is produced by having both of our eyes looking straight ahead and moving together. As the eyes are slightly apart, different views are received from each, although they overlap to a small extent. The brain computes the spatial information from each eye to create an overall image, using previous experience to speed up the processing time and fill in any gaps.

Swapping sides

At a crossover point called the optic chiasm, nerve axons from the left side of each retina join and continue to the left visual cortex, and likewise with nerve axons from the right.

